

Macaulay's contribution [372] was, essentially, the equivalence (i) $\iff$ (ii) of Theorem 8.34. We have combined it with important work by Stanley [512, 513] and by Björner, Frankl & Stanley [93].

The sequences characterized in Theorem 8.34 are called *M-sequences* (dial “M” for “Macaulay”) or “*O*-sequences” (“O” for whatever). They are of fundamental importance, as we will also see in the next section.

Here are a few examples of *M*-sequences, for $d = 3$. The sequence $(1, 3, 3, h_3)$ is an *M*-sequence for $0 \leq h_3 \leq 4$. In fact, part (ii) of Macaulay's theorem suggests that for this we can take the multicomplex

$$\{ \emptyset, 1, 2, 3, 11, 12, 22 \}$$

together with the first h_3 sets from the list

$$F_1(3) = 111, \quad F_2(3) = 112, \quad F_3(3) = 122, \quad F_4(3) = 222.$$

Since $F_5(3) = 113$ contains the submultiset 13 that is not among the 1-faces we listed, we get that $(1, 3, 3, 5)$ is no longer an *M*-sequence. Note that among these, for $h_3 = 1$ we get $(1, 3, 3, 1)$ as an *M*-sequence: this is the h -vector of the boundary of an octahedron; see Example 8.22.

8.6 The g -Theorem and Its Consequences

From the last section, I hope we gathered some intuition for “what an *M*-sequence is.” All kinds of interpretations are useful: so, the best is to alternate between various explanations, between

- the f -vector of a multicomplex,
- the h -vector of a shellable complex, and
- a sequence of nonnegative integers satisfying $\partial^k(h_k) \leq h_{k-1}$.

Here comes one big reason why *M*-sequences are useful. It yields a complete characterization of the f -vectors of simplicial d -polytopes P . What do we know about them so far? Forgetting about $f_d = 1$, we know they can be encoded by their h -vectors $\mathbf{h}(P) = (h_0, h_1, \dots, h_d)$, which satisfy the Dehn-Sommerville equations $h_k = h_{d-k}$ for $0 \leq k \leq d$. Also, we know that $\mathbf{h}(P)$ is an *M*-sequence from Macaulay's Theorem 8.34(iv), which implies the upper bound inequality

$$h_k \leq \left(\binom{h_1}{k} \right) = \binom{h_1 + k - 1}{k}.$$

In quite a daring step, McMullen in 1970 combined all the then available information (including the lower bound theorem by Barnette, see below)

into a conjectured complete characterization [391]. It became known as the g -conjecture, because it referred to the g -vector of the polytope, defined as

$$\mathbf{g}(P) := (g_0, g_1, \dots, g_{\lfloor \frac{d}{2} \rfloor})$$

with $g_0 := h_0 = 1$, and $g_k := h_k - h_{k-1}$ for $1 \leq k \leq \lfloor \frac{d}{2} \rfloor$.

The g -Theorem 8.35. (Billera & Lee [74, 75] and Stanley [511])

A sequence $\mathbf{g} = (g_0, g_1, \dots, g_{\lfloor \frac{d}{2} \rfloor}) \in \mathbb{N}_0^{\lfloor \frac{d}{2} \rfloor + 1}$ is the g -vector of a simplicial d -polytope if and only if it is an M -sequence.

We will not even attempt to prove this (see the notes below): we'll be content with deriving some of its most striking consequences. For this, we use a matrix formulation of the “McMullen correspondence,” due to Björner [84, 86, 90].

In the following, we use the convention $h_{d+1} := 0$ and again define $g_k := h_k - h_{k-1}$ for $0 \leq k \leq d+1$. With this, we find

$$g_{d+1-k} = h_{d+1-k} - h_{d-k} = h_{k-1} - h_k = -g_k \quad \text{for } 1 \leq k \leq d+1.$$

These “Dehn-Sommerville equations for the g -vector” explain why we restrict our attention to g_k for $k \leq \lfloor \frac{d}{2} \rfloor$: we reconstruct $g_k = -g_{d+1-k}$ for $k \geq (d+1) - \lfloor \frac{d}{2} \rfloor$. Careful: there might be one more term in the sequence, namely g_k for $k = \lfloor \frac{d}{2} \rfloor + 1 = d - \lfloor \frac{d}{2} \rfloor$ in the case when d is odd. However, we can ignore this case since this g_k vanishes.

With this, we express the f -vector in terms of the g -vector as follows:

$$\begin{aligned} f_{k-1} &= \sum_{i=0}^d \binom{d-i}{k-i} h_i \\ &= \sum_{i=0}^d \binom{d-i}{k-i} \sum_{j=0}^i g_j \\ &= \sum_{j=0}^{d+1} \sum_{i=j}^d \binom{d-i}{k-i} g_j \\ &= \sum_{j=0}^{d+1} \binom{d+1-j}{d+1-k} g_j \\ &= \sum_{j=0}^{\lfloor \frac{d}{2} \rfloor} g_j \left(\binom{d+1-j}{d+1-k} - \binom{j}{d+1-k} \right). \end{aligned}$$

We take this result and interpret it as a matrix correspondence. For this, we define a coefficient matrix

$$M_d = (m_{jk})_{jk} := \left(\binom{d+1-j}{d+1-k} - \binom{j}{d+1-k} \right)_{0 \leq j \leq \lfloor \frac{d}{2} \rfloor, 0 \leq k \leq d}$$

and use it to restate the g -theorem, as follows.

Theorem 8.36 (The “McMullen correspondence”). [90]

$$\mathbf{g} \longmapsto \mathbf{g} \cdot M_d$$

is a bijection between the M -sequences $\mathbf{g} \in \mathbb{N}_0^{\lfloor \frac{d}{2} \rfloor + 1}$ with $g_1 = n - d - 1$, and the f -vectors $\mathbb{N}_0^{d+1}$ of simplicial d -polytopes with $n = g_1 + d + 1$ vertices.

For example, we compute

$$\begin{aligned} M_1 &= (1 \ 2), & M_2 &= \begin{pmatrix} 1 & 3 & 3 \\ 0 & 1 & 1 \end{pmatrix}, \\ M_3 &= \begin{pmatrix} 1 & 4 & 6 & 4 \\ 0 & 1 & 3 & 2 \end{pmatrix}, & M_4 &= \begin{pmatrix} 1 & 5 & 10 & 10 & 5 \\ 0 & 1 & 4 & 6 & 3 \\ 0 & 0 & 1 & 2 & 1 \end{pmatrix}. \end{aligned}$$

From this we get trivialities for $d \leq 2$. For $d = 3$ we get the f -vectors of simplicial 3-polytopes, which are also easy to get by elementary arguments (Exercise 8.28). However, starting at $d = 4$, we get nontrivial characterizations: so the f -vectors of simplicial 4-polytopes are all the row vectors $\mathbf{f}$ of the following form:

$$\begin{aligned} \mathbf{f}(P_4) &= (1, 5, 10, 10, 5) + g_1(0, 1, 4, 6, 3) + g_2(0, 0, 1, 2, 1) \\ &\text{with } g_1, g_2 \geq 0, \partial^2(g_2) \leq g_1. \end{aligned}$$

The matrices M_d are given explicitly and are not hard to analyze, which allows us to study the f -vectors of simplicial d -polytopes. In particular, one can easily (using well-known recursions, monotonicity properties, and so forth, of binomial coefficients) verify the following simple properties.

Lemma 8.37. *The entries of the matrix M_d are nonnegative integers, with zeroes below the diagonal ($m_{jk} = 0$ for $j > k$), ones on the diagonal ($m_{jj} = 1$ for all j), and larger values above the diagonal ($m_{jk} > 1$ for $j < k$, except for $m_{\frac{d}{2}, d} = 1$ in the case when d is even).*

Instead of a proof, here is the computation of M_d for $d = 7$: we get a (4×7) -matrix, $M_7 =$

$$\begin{aligned} &\begin{pmatrix} \binom{8}{8} - \binom{0}{8} & \binom{8}{7} - \binom{0}{7} & \binom{8}{6} - \binom{0}{6} & \binom{8}{5} - \binom{0}{5} & \binom{8}{4} - \binom{0}{4} & \binom{8}{3} - \binom{0}{3} & \binom{8}{2} - \binom{0}{2} & \binom{8}{1} - \binom{0}{1} \\ \binom{7}{8} - \binom{1}{8} & \binom{7}{7} - \binom{1}{7} & \binom{7}{6} - \binom{1}{6} & \binom{7}{5} - \binom{1}{5} & \binom{7}{4} - \binom{1}{4} & \binom{7}{3} - \binom{1}{3} & \binom{7}{2} - \binom{1}{2} & \binom{7}{1} - \binom{1}{1} \\ \binom{6}{8} - \binom{2}{8} & \binom{6}{7} - \binom{2}{7} & \binom{6}{6} - \binom{2}{6} & \binom{6}{5} - \binom{2}{5} & \binom{6}{4} - \binom{2}{4} & \binom{6}{3} - \binom{2}{3} & \binom{6}{2} - \binom{2}{2} & \binom{6}{1} - \binom{2}{1} \\ \binom{5}{8} - \binom{3}{8} & \binom{5}{7} - \binom{3}{7} & \binom{5}{6} - \binom{3}{6} & \binom{5}{5} - \binom{3}{5} & \binom{5}{4} - \binom{3}{4} & \binom{5}{3} - \binom{3}{3} & \binom{5}{2} - \binom{3}{2} & \binom{5}{1} - \binom{3}{1} \end{pmatrix} \\ &= \begin{pmatrix} 1 - 0 & 8 - 0 & 28 - 0 & 56 - 0 & 70 - 0 & 56 - 0 & 28 - 0 & 8 - 0 \\ 0 - 0 & 1 - 0 & 7 - 0 & 21 - 0 & 35 - 0 & 35 - 0 & 21 - 0 & 7 - 1 \\ 0 - 0 & 0 - 0 & 1 - 0 & 6 - 0 & 15 - 0 & 20 - 0 & 15 - 1 & 6 - 2 \\ 0 - 0 & 0 - 0 & 0 - 0 & 1 - 0 & 5 - 0 & 10 - 1 & 10 - 3 & 5 - 3 \end{pmatrix} \end{aligned}$$

and this means that the f -vectors of simplicial 7-polytopes are exactly the vectors of the form

$$(g_0, g_1, g_2, g_3) \cdot \begin{pmatrix} 1 & 8 & 28 & 56 & 70 & 56 & 28 & 8 \\ 0 & 1 & 7 & 21 & 35 & 35 & 21 & 6 \\ 0 & 0 & 1 & 6 & 15 & 20 & 14 & 4 \\ 0 & 0 & 0 & 1 & 5 & 9 & 7 & 2 \end{pmatrix}$$

for an M -sequence $(g_0, g_1, g_2, g_3) \in \mathbb{N}_0^4$.

From the McMullen correspondence, one gets the upper bound theorem as an immediate consequence, but also the lower bound theorem, which was first proved by Barnette in 1970 [42, 44].

Corollary 8.38 (Upper and lower bound theorem).

We consider simplicial d -polytopes P of fixed dimension d and fixed number of vertices $n = g_1 + d + 1$.

(UBT) The f -vector $\mathbf{f}(P) = \mathbf{g}M_d$ has its componentwise maximum if and only if all the components of $\mathbf{g}$ are maximal, with

$$g_k = \binom{\binom{g_1}{k}}{k} = \binom{g_1 + k - 1}{k} = \binom{n - d + k - 2}{k}.$$

Also, f_{k-1} is maximal if and only if g_i is maximal for all i with $i \leq \min\{k, \lfloor \frac{d}{2} \rfloor\}$.

(LBT) The f -vector $\mathbf{f}(P) = \mathbf{g}M_d$ takes its componentwise minimum if and only if all the components of $\mathbf{g}$ are minimal, that is, if $g_i = 0$ for $i > 1$.

Also, f_{k-1} is minimal if and only if $g_i = 0$ for $2 \leq i \leq \min\{k, \lfloor \frac{d}{2} \rfloor\}$.

An analysis of the matrices M_d can also be applied to the *unimodality conjecture* for convex polytopes: the question of whether for every polytope it is true that the f -vector satisfies

$$1 = f_{-1} \leq f_0 \leq \dots \leq f_{p-1} \leq f_p \geq f_{p+1} \geq \dots \geq f_{d-1} \geq f_d = 1,$$

for some p , that is, the f -vector has to be *unimodal*. It seems that this question was first asked by Motzkin in the late 1950s; see [84].

For this, it is not hard (but a little tedious, perhaps) to check that the rows of M_d are unimodal: they first increase, until they reach a maximum, and then they decrease again. Furthermore, the maximum occurs in columns with indices j between $j = \lfloor \frac{d}{2} \rfloor$ and $j = \lfloor \frac{(3d-1)}{4} \rfloor$. This type of analysis yields the crucial part of the following theorem.

Theorem 8.39. (Björner [84, 90])

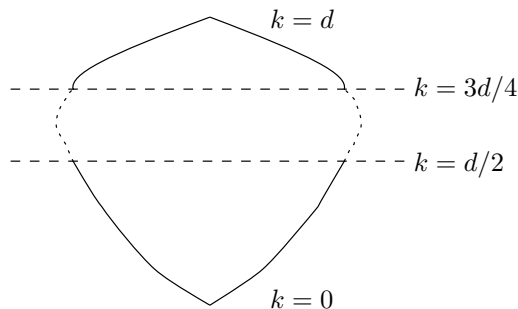
The f -vectors of simplicial d -polytopes with $d \geq 3$ satisfy

$$f_{-1} < f_0 < f_1 < \dots < f_{\lfloor \frac{d}{2} \rfloor - 1} \leq f_{\lfloor \frac{d}{2} \rfloor} \quad \text{and} \quad f_{\lfloor \frac{3(d-1)}{4} \rfloor} > \dots > f_{d-1}.$$

The bounds $\lfloor \frac{d}{2} \rfloor$ and $\lfloor \frac{3(d-1)}{4} \rfloor$ are best possible in the sense that for every p and d with $\lfloor \frac{d}{2} \rfloor \leq p \leq \lfloor \frac{3(d-1)}{4} \rfloor$, there is a simplicial d -polytope whose f -vector “peaks” at p :

$$f_{-1} < f_0 < f_1 < \cdots < f_{p-1} < f_p > f_{p+1} > \cdots > f_{d-1}.$$

Thus, the “shape” of the face lattice of simplicial convex polytopes looks roughly as follows (taking into account also that they are “top heavy”; see Exercise 8.34):



Björner’s theorem implies the unimodality conjecture for simplicial polytopes of dimension $d \leq 10$. With more work, one can get it up to dimension 15 (Björner [84, 90]), and even to dimension 19 (Eckhoff [188]). Surprisingly enough, the unimodality conjecture for simplicial polytopes is false in dimension 20, as was first discovered by Björner [84] and Lee [352, 74].

Examples 8.40. The unimodality conjecture fails for a simplicial polytope of dimension $d = 20$ with the following f -vector, for which $f_{11} > f_{12} < f_{13}$.

f_{-1}	=	1	
f_0	=	4203045807626	
f_1	=	84060916163336	
f_2	=	798578704207074	
f_3	=	4791472253296106	
f_4	=	20363758019368323	
f_5	=	65164051780016980	
f_6	=	162910744316489788	
f_7	=	325834059588060117	
f_8	=	529707205213463823	
f_9	=	709935971390166248	
f_{10}	=	805494832051588614	
f_{11}	=	821976324224631043	/
f_{12}	=	821976324224611712	←
f_{13}	=	822000129478641948	\
f_{14}	=	747383755288236256	
f_{15}	=	546761228419958342	
f_{16}	=	293715859557026466	
f_{17}	=	106920718330384544	
f_{18}	=	23458617733909980	
f_{19}	=	2345861773390998	

To construct such f -vectors we use g -vectors of the form

$$\begin{aligned} g_1 &:= n - d - 1 + r, \\ g_k &:= \binom{n - d - 2 + k}{k} \quad \text{for } k \neq 1. \end{aligned}$$

Now take $d = 20$, $n = 169$, and $r = 4203045807457$, and compute (i.e., let MAPLE compute).

The existence of the corresponding polytope follows from the (necessity part of the) g -theorem. However, the corresponding polytopes $C_d(n)^{<r>}$ are also easy to construct “by hand”: see Exercise 8.32.

If we go a little higher in dimension, then the same construction produces nonunimodal f -vectors for simplicial polytopes with much fewer vertices: so, for $d = 30$, $n = 47$, and $r = 65555$ one obtains a simplicial f -vector with only $f_0 = 65602$ vertices. However, Eckhoff [188] observed that with a more complicated f -vector one can do even better. The simplicial f -vector with the smallest number of vertices he found is

$$\begin{array}{rcl} f_{-1} & = & 1 \\ f_0 & = & 1320 \\ f_1 & = & 869619 \\ f_2 & = & 24650747 \\ f_3 & = & 342491792 \\ f_4 & = & 3070918789 \\ f_5 & = & 19918328394 \\ f_6 & = & 99465082767 \\ f_7 & = & 397591643442 \\ f_8 & = & 1306188319799 \\ f_9 & = & 3593770140180 \\ f_{10} & = & 8397239870111 \\ f_{11} & = & 16843753477928 \\ f_{12} & = & 29259588507633 \\ f_{13} & = & 44370698483306 \\ f_{14} & = & 59263421467414 \\ f_{15} & = & 70604148959649 \\ f_{16} & = & 76609321169592 \\ f_{17} & = & 78245589858777 & / \\ f_{18} & = & 78245589349944 & \longleftarrow \\ f_{19} & = & 78245589350797 & \backslash \\ f_{20} & = & 76598891788386 \\ f_{21} & = & 69592677861523 \\ f_{22} & = & 55485099387534 \\ f_{23} & = & 37137014371927 \\ f_{24} & = & 20144065902012 \\ f_{25} & = & 8558343705069 \\ f_{26} & = & 2730558787586 \\ f_{27} & = & 613985498319 \\ f_{28} & = & 86678396880 \\ f_{29} & = & 5778559792 \end{array}$$

This is obtained from the g -vector

$$\begin{aligned} g_0 &= 1 \\ g_1 &= 1289 \\ g_2 &= 830484 \\ g_i &= \binom{i+18}{i} + \binom{i+16}{i-1} \quad \text{for } 3 \leq i \leq 14 \\ g_{15} &= 1252344784 \end{aligned}$$

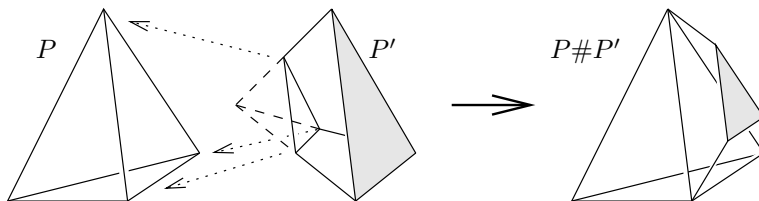
The existence of a corresponding simplicial polytope follows from the sufficiency part of the g -Theorem 8.32, if we verify that these g_i form an M -sequence, and this is easy — do it!

From these f -vectors of “large” polytopes, you can perhaps get a more realistic feeling for what f -vectors look like “in practice.” Observe how the monotonicity statements of Björner’s Theorem 8.39 still hold. I guess the more general moral is that you shouldn’t rely too much on intuition from 3- and 4-polytopes when you want to get a feel for the behavior of “typical” simplicial polytopes.

Furthermore, in contrast to all the detailed (essentially complete, by the g -Theorem 8.35!) information known about f -vectors of simplicial polytopes, we do not know much about nonsimplicial polytopes. Our knowledge is not even complete for 4-polytopes (Problem 8.29*). In Problems 8.33* and 8.35* we ask basic questions about the f -vectors of general polytopes. Here we end the chapter with a construction — due to Eckhoff [188] — that “easily” produces nonunimodal f -vectors in low dimensions.

Example 8.41. (Eckhoff [188])

Let P be a simplicial polytope, and P' a simple polytope, both of dimension d . Now we “cut off” one vertex from P' ; then, after a projective transformation, we can “glue” the rest of P' onto a facet of P , to obtain the *connected sum* $P \# P'$. Instead of formal details for this construction, we just provide a sketch of a 3-dimensional connected sum.



Thus, the sum $\Delta_3 \# \Delta_3$ is combinatorially equivalent to the capped prism.

Now if P has the f -vector

$$\mathbf{f}(P) = (1, f_0, f_1, \dots, f_{d-2}, f_{d-1})$$

and P' has the f -vector

$$\mathbf{f}(P') = (1, f'_0, f'_1, \dots, f'_{d-2}, f'_{d-1})$$

then $P \# P'$ will have the f -vector

$$f(P \# P') = (1, f_0 + f'_0 - 1, f_1 + f'_1, \dots, f_{d-2} + f'_{d-2}, f_{d-1} + f'_{d-1} - 1)$$

— this is just the sum of the f -vectors, except that 1 has been subtracted in the dimensions -1 , 0 and $d-1$, corresponding to the vertex of P' and the facet of P that were deleted in the construction.

Now if $P = C_d(n)$ is a cyclic d -polytope with many vertices, then its f -vector peaks in dimension $\lfloor \frac{3(d-1)}{4} \rfloor$, and the f -vector of its polar peaks at in dimension $\lceil \frac{d-1}{4} \rceil$. This suggests that, if d and n are large enough, then the f -vector of $C_d(n) \# C_d(n)^\Delta$ cannot be unimodal. For example, straightforward computations, for $d = 8$ and $n = 25$, yield

$$\begin{aligned} f(C_8(25) \# C_8(25)^\Delta) &= \\ &= (1, 7149, 28800, 46800, 46400, 46400, 46800, 28800, 7149). \end{aligned}$$

Similarly, for $d = 9$ and $n = 18$ one gets

$$\begin{aligned} f(C_9(18) \# C_9(18)^\Delta) &= \\ &= (1, 1447, 6588, 12984, 15618, 15552, 15618, 12984, 6588, 1447). \end{aligned}$$

These polytopes, in dimension 8, and with less than 1500 vertices, you might even consider as “small” (if you compare them to our previous, simplicial examples).

Notes

Sections 8.1 and 8.2. Schläfli [473] had made a shellability assumption for his 1852 proof of the d -dimensional Euler-Poincaré formula, but did not specify the exact condition he needed. Thus the theory of shellability got its basis with the paper by Bruggesser & Mani [139], published in 1971, in which they first defined the concept. Bruggesser and Mani write in their introduction: “We were surprised to find that Schläfli’s assumption can be justified in an almost trivial manner” [139, p. 197].

Since then, shelling has become a very basic and useful technique with many (geometric and combinatorial) applications. We refer to Danaraj & Klee [172], Björner & Wachs [98], Björner et al. [96, Sect. 4.7], and in particular to Björner [87] for further reading and references.

A nonshellable triangulation of a tetrahedron (with 14 vertices, 41 facets, all vertices on the boundary) was constructed by Rudin [468] in 1958. This discouraged geometers from trying to prove that the boundary complex of every polytope is shellable. Rudin’s ball can even be brought into convex position [162, p. 305], so it can be considered as a nonshellable triangulation of a 3-polytope without new vertices.